

AFL Data Dictionary + Target Definitions

1. Data Dictionary

Dataset	Grain / One Row Represents	Key Fields	Important Variables
Players (afl_players_info_raw.csv)	One row per player	id	player_name, born_date, debut_date, height, weight, player_teams
Player Match Stats (afl_players_round_by_round_stats_raw.csv)	One row per player per match	player_id, year, round	team, opponent, match_date, result, kicks, marks, handballs, disposals, goals, tackles, clearances, brownlow_votes, fantasy_points, score, margin
Player Seasonal Stats (afl_players_seasonal_stats_raw.csv)	One row per player/team/season/finals status	player_id, year, team, is_finals	games_played, disposals, goals, brownlow_votes, total_fantasy_points, avg_disposals, avg_goals, avg_fantasy_points
Team Matches (team_matches_home_away_raw.csv)	One row per team perspective of a match	team_name, opponent, year, round, match_date	home_away, team_score, opponent_score, result, margin, venue, crowd, team_goals_kick

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Relationships: players.id → player_match.player_id and players.id → player_season.player_id. Team and player match records are connected using normalized team, opponent, year, round, and match_date. There is no explicit match_id, so a composite match key is used.

Important limitations: The available data does not contain explicit player position, weather, or direct travel-distance variables. Therefore, position-based and weather/travel-specific analysis cannot be performed directly.

2. Prediction Target Definitions

Target	Definition	Formula	Level
team_win_flag	Whether a team wins the match	Win = 1, Loss = 0; draws handled separately	Team-match
team_margin	Points scored by the team relative to opponent	team_score – opponent_score	Team-match
top_disposal_getter	Player with the highest disposals in a match/period	argmax(disposals)	Player-match
top_goal_kicker	Player with the highest goals in a match/period	argmax(goals)	Player-match
fantasy_value_score	Player fantasy output normalized by games played	fantasy_points / games_played	Player-season
brownlow_style_score	Player performance measure based on Brownlow votes	brownlow_votes aggregated over the selected period	Player-season

Primary Modeling Targets

Match prediction: team_win_flag is the primary classification target because the main objective is to predict the match winner. team_margin is the secondary regression target because predicting the scoring margin provides a more detailed measure of match outcome.

Player prediction: top_goal_kicker and top_disposal_getter are the primary player-level targets. fantasy_value_score provides an additional composite performance target.

Leakage Rule

All predictive features must use information available **before the target match**. Rolling averages, win streaks, head-to-head statistics, rest days, and ladder position are therefore calculated using only previous matches. Future-match information must never be included in the feature values.